CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Real Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The largest number dividing 113, 197 and 281 leaving remainder 1 in each case is: 1

(a) 28 **(b)** 56 **(c)** 6 **(d)** 14

2. The smallest positive integer by which 28 must be multiplied to obtain a perfect square is: 1

(a) 14 **(b)** 28 **(c)** 7 **(d)** 2

3. Which of these is a rational number? 1

(a) $7\sqrt{10}$ **(b)** $\sqrt{7}+\sqrt{8}$ **(c)** $\sqrt{9}$ **(d)** $\dfrac{\sqrt{147}}{\sqrt{3}}$

4. A number ends with exactly 5 zeros if its prime factorisation (other factor 3) is: 1

(a) $2^4 \times 5^5 \times 3$ **(b)** $2^6 \times 5^4 \times 3$ **(c)** $3^5 \times 7$ **(d)** $2^6 \times 5^5 \times 3$

5. The LCM of 28, 42 and 63 is: 1

(a) 252 **(b)** 126 **(c)** 2646 **(d)** 1176

6. The least number that is divisible by every number from 1 to 10 is: 1

(a) 2520 **(b)** 5040 **(c)** 1260 **(d)** 1814400

7. The largest number dividing 59, 101 and 143 leaving remainder 3 in each case is: 1

(a) 2 **(b)** 7 **(c)** 14 **(d)** 28

8. The smallest positive integer by which 90 must be multiplied to obtain a perfect square is: 1

(a) 2 **(b)** 30 **(c)** 90 **(d)** 10

9. **Assertion (A):** If two numbers are co-prime, their LCM equals their product. 1
Reason (R): The HCF of two co-prime numbers is 1.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** The number 6^n can end with the digit 0 for some natural number n. 1
Reason (R): A number ends with 0 only if it is divisible by both 2 and 5.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Prove that $4\sqrt{11}$ is irrational, given that $\sqrt{11}$ is irrational. 2

12. Two numbers have HCF 18 and are in the ratio $2 : 7$. Find the numbers and their LCM. 2

13. Find the smallest positive integer that leaves remainders 9 and 17 when divided by 16 and 24 respectively. 2

14. Prove that $6\sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A, B and C take 22, 33 and 44 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

16. Find the greatest number that divides 1815 and 2415 leaving remainder 3 in each case. 3

17. Is $9 \times 10 \times 11 + 11$ a composite number? Justify. 3

OR

A, B and C take 14, 35 and 49 minutes per round of a circular track. They start together. After how many minutes do they meet at the start again, and how many rounds has each completed? 3

18. Find the greatest number that divides 3011 and 3431 leaving remainder 11 in each case. 3

OR

Is $10 \times 11 \times 12 + 12$ a composite number? Justify. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A teacher gives the class the numbers $\frac{13}{125}$, $\frac{17}{18}$ and $\frac{29}{343}$ and asks students to decide the nature of their decimal expansions without dividing. 5

(i) Write the denominator of $\frac{13}{125}$ in prime factors.

(ii) State the condition on the denominator for a rational number to have a terminating decimal expansion.

(iii) Which of the three numbers have terminating expansions?

(iv) Find the decimal expansion of $\frac{13}{125}$ and state, with reason, whether $\frac{17}{18}$ terminates.

20. Three sets of English, Hindi and Mathematics books have 108, 162 and 270 books. They are stacked so that each stack has the same number of books of one subject and all stacks have the same height. Find the minimum number of stacks. 5

OR

Prove that $\frac{1}{\sqrt{13}}$ is irrational and hence that $\frac{6}{\sqrt{13}}$ is irrational. 5

END OF QUESTION PAPER — Total: 40 Marks